

The *sigaction* structure includes the following members:

```
struct sigaction {
    void    (*sa_handler)();
    void    (*sa_sigaction)(int, siginfo_t *, void
*);
    sigset_t sa_mask;
    int     sa_flags;
};
```

The function could be used to get and modify the disposition for a specified signal.

The *sigprocmask()* API is used to block/unblock a specified signal and *sigsuspend()* is used to suspend the process till it receives a specified signal or the process gets killed.

The *sigaddset()*, *sigdelset()*, *sigemptyset()*, *sigfillset()*, *sigismember()* are the auxiliary functions available and should be used to operate on *sigset_t*.

A piece of code to catch SIGINT

Here is a simple piece of code to catch *SIGINT*:

```
/****** SigInt.c
******/
#include <signal.h>
#include <stdio.h>
#include <unistd.h>

void SigIntHandler(int sig)
{
    printf("Received signal %d\n", sig);
}

int main()
{
    struct sigaction act;
    int count = 5;

    act.sa_handler = SigIntHandler;
    sigemptyset(&act.sa_mask);
    act.sa_flags = 0;

    sigaction(SIGINT, &act, 0);

    while(count--) {
        printf("Looping!\n");
        sleep(5);
    }
}
```

```
}
}
```

Here we compile and run the code:

```
# gcc SigInt.c
# ./a.out
Looping!
Looping!
Received signal 2
Looping!
Looping!
Received signal 2
Looping!
#
```

Watch for some pitfalls

The following are some points to be considered during the design time:

1. A process might wait for an indefinite period of time when it invokes *sigsuspend()* but does not receive the desired signal.
2. Due to a timing mismatch, a process might wait for a signal that has already occurred.
3. The child process inherits the same signal handlers from the parent after *fork()*.
4. The final and most important point is that the signal handlers should be short and reentrant.

A nice article about signals and reentrancy is available at www.ibm.com/developerworks/linux/library/l-reent.html

Here onwards

This article talks about the basics of the 'signals' framework on Linux. Some more advanced and interesting stuff is available in a *Linux Journal* article on signals: www.linuxjournal.com/article/3985. 

By: Nilesh Govande

The author is a Linux enthusiast and can be contacted at nileshgovande@yahoo.com. His areas of interest include Linux systems software, application development and virtualisation. He is currently working with the LSI Research & Development Centre, Pune.




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